Few-Shot Hyperspectral Image Classification With Unknown Classes Using Multitask Deep Learning

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***Abstract*— Current hyperspectral image classification assumes that a predefined classification system is closed and complete, and there are no unknown or novel classes in the unseen data. However, this assumption may be too strict for the real world. Often, novel classes are overlooked when the classification system is constructed. The closed nature forces a model to assign a label given a new sample and may lead to overestimation of known land covers (e.g., crop area). To tackle this issue, we propose a multitask deep learning method that simultaneously conducts classification and reconstruction in the open world (named MDL4OW) where unknown classes may exist. The reconstructed data are compared with the original data; those failing to be reconstructed are considered unknown based on the assumption that they are not well represented in the latent features due to the lack of labels. A threshold needs to be defined to separate the unknown and known classes; we propose two strategies based on the extreme value theory for few- and many-shot scenarios. The proposed method was tested on real-world hyperspectral images; state-of-the-art results were achieved, e.g., improving the overall accuracy by 4.94% for the Salinas data. By considering the existence of unknown classes in the open world, our method achieved more accurate hyperspectral image classification, especially under the few-shot context.**

***Index Terms*— Classification with unknown classes, convolutional neural network (CNN), deep learning, hyperspectral image classification, multitask learning, open-set recognition.**

I. Introduction

THE world's urbanization has rapidly developed since the 1980s. With the growing population, a large number of human beings living on this small planet becomes a crucial problem. Sufficient food supplies are needed to feed human beings, but sufficient land resources for daily life activities are also needed [1]. Ongoing urbanization transforms natural environments into cities and increases the fraction of the impervious surface, eliminating rainwater infiltration and increasing the urban temperature. Planning land resource usage wisely is the key to maintaining the world's sustainability, and this requires timely and accurate land cover monitoring.

Existing methods for land cover mapping include land survey and classification based on satellite and airborne remote sensing images. A land survey is impractical for large area monitoring as it is labor-intensive and time-consuming. Classifying each pixel of a satellite image is a more popular way. Two existing semiautomatic methods can be applied: semantic segmentation and pixelwise classification. Semantic segmentation considers the entire image as a sample instance and directly outputs the segmentation result [2]; it requires a large set of fully annotated images [3]. This approach is not practical for detailed land cover mapping as: 1) the land cover classification system is very complex and 2) the satellite data are quite diverse. Especially, depending on the application, a remote sensing image can be either mapped into three to four classes (impervious, vegetation, water, and sometimes barren) or more than ten fine-grained classes [4]–[6]. The land cover classification system is multilevel. On the other hand, the sources of remote sensing images are diverse, making it difficult to reuse previous data sets. For optical remote sensing, the number of spectral bands ranges from one to 100. The diversity increases if nonoptical data [e.g., polarimetric synthetic aperture radar (PolSAR)] are included. As a result, pixelwise classification is more popular for land cover mapping.

Remotely sensed hyperspectral images, due to their rich spectral information, are of great use for land cover classification. The physical theory for hyperspectral image classification is that each class can be separated based on its spectral profiles. Due to its great potential, hyperspectral image classification has gone through rapid development in the last two decades [7]–[9]. In the early 2000s, hyperspectral images were solely considered as sequential data, as shown in the classic 2-D storage format (#sample × #feature). At this stage, support vector machines (SVMs) were the mainstream classifiers [10]–[12]. SVMs with kernels can project hyperspectral data to a high-dimensional hyperplane, where the training data should be highly separated based on their class belongings. In the late 2000s, with the popularity of graph models (e.g., Markov random fields [13]), morphological profiles [14], and other spatial filters, including edge-preserving filters and Gabor filters [15], [16], spectral–spatial classification of hyperspectral data became a popular topic and influenced the last ten years [7]. At the same time, many advanced algorithms tackling the limited availability of training samples were proposed, including active learning [17], [18], semisupervised learning [19], [20], graph models [21], sparse representa-

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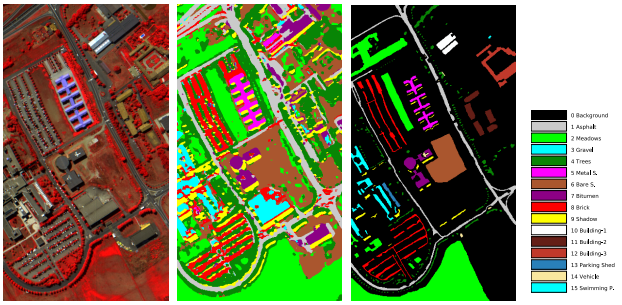
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tion [22], [23], and domain adaptation [24]. Since the classification performance also depends on the dimension due to the curse of dimensionality, band selection and dimension reduction techniques are widely used in hyperspectral image classification [25]–[27].

In the deep learning era, the novel convolutional neural networks (CNNs), recurrent neural networks (NNs), and graph convolutional networks started to dominate the classification of hyperspectral and remote sensing data [9], [28]–[34]. Manual feature engineering was replaced by automatic feature learning by deep networks. Now, hyperspectral image classification entered the stage with 99% classification accuracy with sufficient training samples [35]. Besides, many works have focused on improving the few-shot classification accuracy (classification with limited samples), including deep few-shot learning (DFSL) with NNs and SVM [36], spatial–spectral relation network (SS-RN) [37], and deep relation network [38]. However, the high accuracy is achieved under the assumption that there is no unknown class in the unseen data. If an instance does not belong to any of the known classes, this instance will not harm the classification accuracy from the current evaluation system, as such instance will not be considered. However, the existence of unknown or novel classes will lead to false positives and, thereby, reduce the precision of a model. If many instances belong to the unknown class are classified as a known class, the number of the known classes will be largely overestimated. Unfortunately, in the real world, it is difficult, if not impossible, to define all potential classes. As mentioned, a sample collection based on a field survey can only cover a small portion of the study area due to budget limits. For example, [39] collected ground truth data along with the major road network, and those buried far from the road network were inevitably neglected. Therefore, a machine learning model trained on the collected samples might face some instances (the unknown) that cannot be classified into one of the known classes.

The influence of unknown classes, although dependent on the area of an image, may even occur in a very small study region. The University of Pavia data set, one of the most popular hyperspectral benchmarks, records nine land cover classes in a region of 610×340 with 1.3 m ground sampling distance (GSD), approximately 793×442 m^2. This data set is small and barely covers a campus. However, there is a swimming pool that should not belong to any of the nine classes. We show the false-color map of the University of Pavia data in Fig. 1, where the swimming pool is located on the upper right side of the image. In addition, on the right side of the image, three groups of buildings (building-1, building-2, and building-3) are not annotated in the original reference map. These buildings are often misclassified as natural land covers, such as bare soil [35], [40]–[42] and meadows [43], or other unrelated materials, such as asphalt [44], [45].

The assumption that all test data are known in machine learning is called a closed world or closed set [46], [47]. In the closed world, given an unknown instance that cannot be classified as one of the known classes, a model will eventually assign a known class label to the unknown instance because it cannot identify it as unknown. In this case, if the model is used to map certain crop types, it is inevitable to overestimate their area and, therefore, the total amount of food supplies.



(a) (b) (c)

Fig. 1. Example of the University of Pavia's classification map (nine classes, OA = 98%). The original ground truth covers only nine classes. We manually annotated six additional classes, which were unknown to the classifier. Note building-1, building-2, and building-3 on the right are misclassified as a mix of bare soil and other materials. The swimming pool, helicopters (vehicle), and the parking shed are all misclassified as natural land covers. (a) Image. (b) Classification. (c) Ground truth.

The closed-world setting is not enough for accurate hyperspectral image classification. The machine needs to tell the unknown from known classes. This is impossible for many algorithms, as they assume a closed-world setting. For example, the deep NN is limited by the SoftMax function, which constrains the output probabilities to have a summation of one, and each is within zero to one. In this scenario, one of the known classes must be assigned for the input instance. A simple solution is using confidence thresholding, rejecting those with low classification confidence. However, low classification confidence is not the assurance of the unknown; instead, it is uncertain [46]. An uncertain instance may be a valuable sample close to the classification hyperplane's boundary for active learning, instead of the unknown [48].

To tackle the open world problem, some methods were proposed for daily image recognition. First named open-set recognition by [46], they introduced a one-versus-set machine to empower SVM with the capability to identify the unknown. The OpenMax method is the first one tackling this problem under the context of deep learning [49], which enables the network to estimate the probability of one instance being the unknown class. It used the activation vectors (the final layer's output features before SoftMax) to estimate the distance distribution of each class and then applied the extreme value theory (EVT) to recalibrate the activation vector with $C + 1$ classes, where the added class was the unknown. Neal *et al.* [50] proposed to use counterfactual images generated from generative adversarial networks as the unknown to train the classifier. Dhamija *et al.* [51] argued to use the background classes to reduce the network agnostophobia. Furthermore, [52] added an auxiliary task to enhance the latent features for the OpenMax method. This enhanced method for open-set recognition, named Classification-Reconstruction learning for Open-Set Recognition (CROSR), was tested with anomaly detection algorithms, i.e., IsoForest and one-class SVM, in a classwise fashion, to exploit the latent features effectively. The method achieved the current state of the art

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in open-set recognition. However, like other methods, a large number of training samples were still needed to estimate the centroid of each class, limiting its performance in hyperspectral image classification.

The abovementioned methods for daily image recognition are mostly centroid-based, where the unknown score is calculated in a classwise fashion, which limits their usage under the few-shot context. For example, the Canadian Institute For Advanced Research (CIFAR) data set consists of 600 samples per class [53], and the fashion-Modified National Institute of Standards and Technology (MNIST) consists of 7000 samples per class [54]. In hyperspectral image classification, the size of the training set is significantly smaller than the daily image recognition as collecting ground truth data is labor-intensive and time-consuming. Sometimes, only five to 20 instances per class are available for training [36], [55], [56], limiting the usage of these open-set methods for hyperspectral image classification. Another difference between daily image recognition and hyperspectral image classification is that the number of potential classes is smaller in the latter, as hyperspectral images are taken from the same direction (overhead), whereas daily images are more complex. For hyperspectral images, it is rare to have more than 20 classes in a data set, but, for daily image recognition, the number of classes already exceeds 1000, e.g., ImageNet [57]. As a direct result of fewer potential classes, the openness (unknown degree) of hyperspectral image classification is significantly smaller than daily image recognition. Often, the unknown or novel instances count for only a small portion of the study area, which is called the tail class in technical terms, e.g., swimming pools and vehicles. The low openness significantly affects the performance of current open-set techniques as they are designed to face a larger open world [58]. In summary (as shown in Table I), due to the three notable differences between hyperspectral image classification and daily image open-set recognition (few-shot, fewer classes, and low openness), it is difficult to directly use open-set techniques developed for daily image recognition in hyperspectral image classification.

In this article, tackling the few-shot nature, we propose a novel multitask deep learning method for hyperspectral image classification with unknown classes in the open world (MDL4OW), which is free from estimating the class centroid in a classwise fashion. The proposed method utilizes multitask learning to conduct classification and reconstruction simultaneously. The unknown score is calculated by comparing the original and reconstructed data, whereas the unknown should be poorly reconstructed due to the lack of labels. In this way, the proposed MDL4OW does not need to calculate the mean activation vectors or latent features in a classwise fashion. Therefore, it is suitable for few-shot hyperspectral image classification. The major contributions of this study are summarized as follows.

1) We propose a novel multitask deep learning method named MDL4OW for hyperspectral image classification with unknown classes. The proposed method can identify unknown classes and significantly improve the classification accuracy.
2) Instead of estimating the unknown score in a classwise manner using centroid-based methods, the proposed method can estimate the unknown score with all data using the statistical model EVT. Therefore, the proposed method outperformed the state-of-the-art open-set technique for few-shot hyperspectral image classification.
3) A new evaluation metric, the mapping error, is proposed to evaluate the accuracy of hyperspectral image classification with unknown classes. This metric is especially sensitive to imbalanced classification, which is often the case in hyperspectral images.

The remainder of this article is organized as follows. In Section II, we describe the methodology in detail. In Section III, we introduce the evaluation metrics and the experimental setup. In Section IV, we discuss the experimental results obtained on real-world data sets and provide analyses. Finally, we present our conclusions in Section V.

II. Methodology

In this section, we review a single-task classification network's components, including a feature extractor (encoder) and a classifier. Then, we describe the proposed multitask network in detail, including an encoder–decoder structure that is used to reconstruct the input instance. Based on the assumption that instances with labels from the classification system can be better reconstructed than the unknown, those with a large reconstructed loss should be rejected and considered as unknown. Finally, EVT is used to separate the unknown from known classes.

A. Deep CNNs for Classification

The proposed network with the flowchart is shown in Fig. 2, which consists of two residual learning blocks [59] and a global average pooling layer [60] as the encoder, a fully connected layer with the SoftMax function as the classifier, and a decoder with deconvolutional layers [61]. The classification part is a modified version of HResNet [43] when adding the global average pooling layer.

The powerful ability of deep CNNs for hyperspectral image classification tasks lies in the usage of convolutional layers to extract spectral–spatial information [62]. Considering X as the sample instance space, where each instance $x \in X$, given a limited training set with index k consisting of only a few samples (x^k, l^k), where $l^k \in L = \{1, \ldots, |C|\}$ is the label index for x^k, the multilevel convolutional layers, along with batch normalization [63] and the rectified linear unit (ReLU) [64], serve as a good encoder $\phi(\cdot)$ to extract the spectral–spatial features x_ϕ as the representation of the sample instance

$$x_\phi = \phi(x). \tag{1}$$

In the encoder, the output of a convolutional layer $\phi_{\text{conv}}(\cdot)$ can be simplified as

$$x_{\phi_{\text{conv}}} = \phi_{\text{conv}}(x). \tag{2}$$

Then, the classifier $f(\cdot)$ takes the output vector x_ϕ from the feature extractor $\phi(\cdot)$ as its input. In a pure deep learning

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TABLE I
COMPARISON AMONG HYPERSPECTRAL IMAGE CLASSIFICATION WITH UNKNOWN CLASSES AND THE EXISTING TASKS

Task	Few-shot	Unknown classes	Openness	#Class	#Channel
Daily photo recognition	Rare (often >500)	Yes	High	100+	1-4
Hyperspectral image classification	Often (<50)	No	0	20-	Up to 300+
Hyperspectral image classification with unknown classes	Often (<50)	Yes	Low	20-	Up to 300+

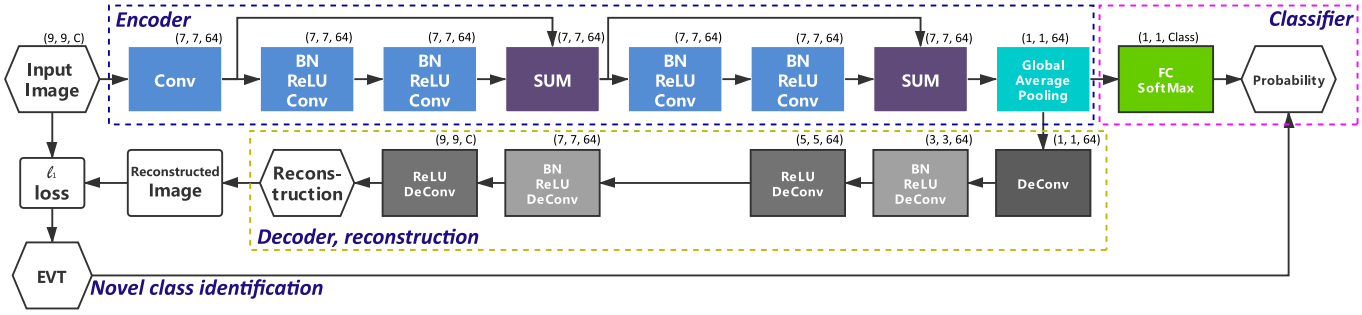


Fig. 2. Flowchart of the proposed method. The multitask network used here takes an input of 9 × 9 × Channel. The first part of the network is an encoder/feature extractor with two residual units and a global average pooling layer. After extracting the latent features, a fully connected layer with the softmax function serves as the classifier and outputs the probability to the known classes. The reconstruction task here uses the deconvolutional layers to increase the spatial dimension of the latent features gradually. The output of the reconstruction task is a 9 × 9 × Channel instance, which should be similar to the input data by minimizing the ℓ_1 loss. The reconstruction loss is modeled by EVT to separate the known and unknown classes.

scenario, the fully connected layer with the SoftMax activation function serves as the classifier $f(\cdot)$ and gives the probability $P(y = j|x_\phi)$ of the jth category

$$P(y = j|x_\phi) = \frac{\exp\left(x_\phi^T w_j + b_j\right)}{\sum_{c=1}^{C} \exp\left(x_\phi^T w_c + b_c\right)} \tag{3}$$

where w_j is the weight vector of the jth neuron in the fully connected layer, b_j is a bias element corresponding to the jth neural, and C is the number of the category. The classification task is to find the optimal parameters for the network by minimizing the cross-entropy loss function ℓ_c

$$\ell_c(y, \hat{y}) = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{4}$$

where C is the number of predefined classes, y is the ground truth label, and $\hat{y}$ is the predicted label.

As mentioned earlier, the SoftMax function transforms the score vector into a probability vector, and the class with the largest probability is considered the predicted class. A naive solution to identify unknown classes is considering those instances with the largest probability smaller than 0.5 as unknown, which is one of the baselines in the experiments (SoftMax with threshold = 0.5).

B. Reconstruction via Multitask Learning

So far, the feature extractor and classifier together form a decent machine for classification, but the network cannot still identify the unknown. To empower the network with this ability, we add a reconstruction task

$$\hat{x} = f_r(x_\phi) \tag{5}$$

where $\hat{x}$ is the reconstructed instance, $f_r(\cdot)$ is the reconstruction function or named the decoder, and x_ϕ is the output latent features from the encoder $\phi(\cdot)$. Here, we use ℓ_1 distance as the reconstruction loss

$$\ell_r(x, \hat{x}) = \|x - \hat{x}\|_1. \tag{6}$$

In the training phase for the multitask network, we minimize the total loss via backpropagation

$$\min_{\phi(\cdot), f_c(\cdot), f_r(\cdot)} \lambda_c \ell_c(y, \hat{y}) + \lambda_r \ell_r(x, \hat{x}) \tag{7}$$

where λ_c and λ_r are the weights to control the loss influence on the multitask network for ℓ_c and ℓ_r, respectively.

In the decoder, the key element is the deconvolutional layer, also called the transposed convolutional layer. It can be considered as the inverse of a convolutional layer, denoted as $\phi^\dagger_{\text{conv}}(\cdot)$, resulting

$$x = \phi^\dagger_{\text{conv}}(\phi_{\text{conv}}(x)). \tag{8}$$

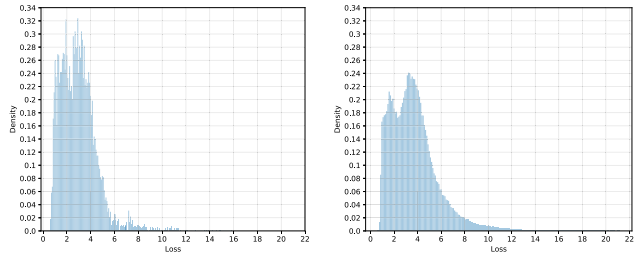
Using the deconvolutional layer, we can gradually increase the $1 \times 1 \times$ channel latent features to patch-based hyperspectral sample instances. As shown in the flowchart (Fig. 2), the reconstruction branch of the proposed method consists of a total of five deconvolutional layers, where all increase the spatial dimension of the instance except for the first one.

C. Threshold Setting With EVT

After obtaining the reconstruction losses (an example is shown in Fig. 3), where a larger loss indicates that the deep learning model is not sufficiently optimized for the instance; the tail part of the loss distribution should be considered as unknown to the model. Here, we adopt the EVT to find the unknown classes by modeling distribution of the tail part.

The EVT suggests that the tail should be a Weibull distribution [65]. For a large class of distributions V and a

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(a) (b)

Fig. 3. Histogram samples of the reconstruction loss. (a) Training data. (b) Test data.

large enough threshold w, with $\{V_1, \ldots, V_n\}$, n independent and identically distributed samples, the cumulative distribution function can be approximated by a generalized pareto distribution (GPD)

$$P(v - w \le v | v > w) = \frac{F_V(w+v) - F_V(v)}{1 - F_V(w)} \tag{9}$$

$$= G_{\xi,\mu}(v) \tag{10}$$

where

$$G_{\xi,\mu}(v) = \begin{cases} 1 - \left(1 + \dfrac{\xi \cdot v}{\mu}\right)^{-\frac{1}{\xi}}, & \xi \neq 0 \\ 1 - e^{-\frac{v}{\mu}}, & \xi = 0 \end{cases} \tag{11}$$

where $\mu > 0$, and $v \ge 0$ when $\xi \ge 0$, and $0 \le v \le -(\mu/\xi)$ when $\xi < 0$. The parameters ξ and μ can be estimated from the given tail data [66]. Here, v is the reconstruction loss, and $G_{\xi,\mu}(\cdot)$ is the cumulative distribution function of GPD. The pseudocode for identifying the unknown class after obtaining the trained network is shown in Algorithm 1.

To model the tail distribution, one defines the tail size τ and a threshold value z to decide the boundary score between known and novel. The threshold is often set as 0.5 [49], [52], whereas the τ value should be at least two to model a distribution. The proposed method MDL4OW is free from class centroid estimation, which enhances its performance under the few-shot context. For comparison purposes, the classwise version MDL4OW/C is also applied. The two methods are similar, except that MDL4OW is applied in a global fashion and MDL4OW/C is applied in a classwise fashion. Details of the threshold setting in the experiments are shown in Section III-C.

Algorithm 1 MDL4OW

Input: $X, \phi(\cdot), f(\cdot), f_r(\cdot), G_{\xi,\mu}(v), \tau, z$
Output: y_{pred}

```
Latent features: X_φ = φ(X)
Predicted known class: y_pred = f(X_φ)
Reconstruction: X̂ = f_r(X_φ)
Reconstruction error: v = ||X − X̂||_1
Compute EVT model G_{ξ,μ}(v) based on v, τ
if G_{ξ,μ}(v) < z then
    y_pred = y_pred
else
    y_pred = unknown
end if
return y_pred
```

III. Data Sets and Experimental Setup

A. Data Sets

1) University of Pavia: The first data set used in the experiment is the previously mentioned benchmark, the University of Pavia. This image was captured by the reflective optics system imaging spectrometer (ROSIS) sensor over the University of Pavia campus in 2001. After the removal of 12 bands affected by noise and water absorption, it contains 103 bands with a scene size of 610 × 340 in a GSD of 1.3 *m*. The standard reference map contains nine classes. As mentioned in Section I, some of the unannotated buildings have a very different spectral profile compared with the known land covers.

TABLE II
University of Pavia

Index	Name	#sample
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Metal sheet	1345
6	Bare soil	5029
7	Bitumen	1330
8	Brick	3682
9	Shadow	947
	Novel	5163
	Total	47939

TABLE III
Salinas Valley

Index	Name	#sample
1	Weeds-1	2009
2	Weeds-2	3726
3	Fallow	1976
4	Fallow-P	1394
5	Fallow-S	2678
6	Stubble	3959
7	Celery	3579
8	Grapes	11271
9	Soil	6203
10	Corn	3278
11	Lettuce-4wk	1068
12	Lettuce-5wk	1927
13	Lettuce-6wk	916
14	Lettuce-7wk	1070
15	Vinyard-U	7268
16	Vinyard-T	1807
	Novel	5613
	Total	59742

We manually annotated six additional classes, as shown in the reference map in Fig. 4. The number of supporting reference pixels is available in Table II.

2) Salinas Valley: The Salinas Valley data set was used in the experiment to evaluate the proposed method in fine-grained crop mapping using hyperspectral data. This image was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Salinas Valley, CA, USA, with

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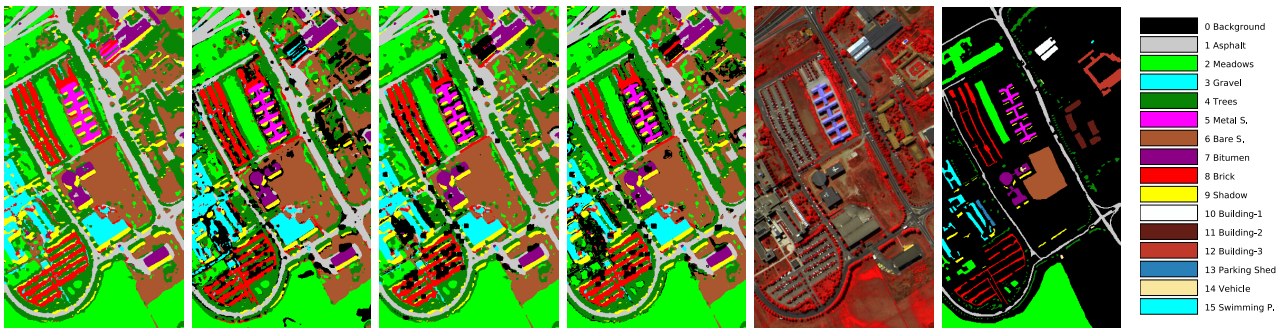


Fig. 4. Classification maps of the University of Pavia. Known classes: 1–9; unknown classes: 10–15. In the classification maps, those masked with black color are identified as unknown by the proposed method. (a) Close. (b) CROSR. (c) MDL4OW. (d) MDL4OW/C. (e) Image. (f) Reference.

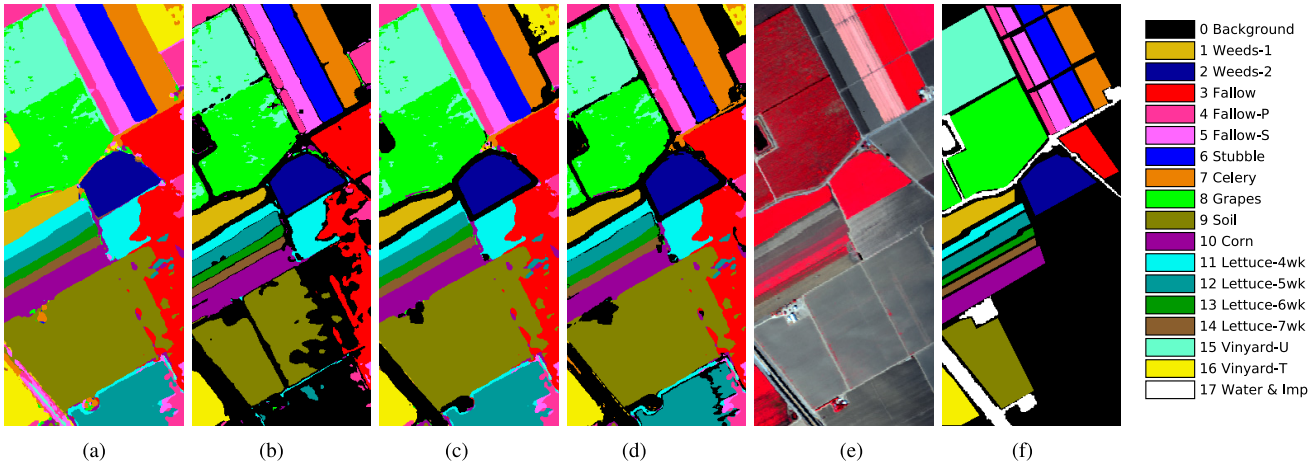


Fig. 5. Classification maps of the Salinas Valley. Known classes: 1–16; unknown classes: 17. In the classification maps, those masked with black color are identified as unknown by the proposed method. (a) Close. (b) CROSR. (c) MDL4OW. (d) MDL4OW/C. (e) Image. (f) Reference.

a GSD of 3.7 m. The region covers an area of 512 × 217 pixels with 204 bands after 20 water absorption bands were discarded. The data were at-sensor radiance data without atmospheric correction. A total of 16 agricultural land covers are identified in the image. However, some of the man-made materials, such as buildings, are not represented in the classification system and should not be classified as any of the known crops. We manually annotated these man-made materials and a water pool as the unknown, as shown in Fig. 5 (reference map). Details of the data set can be found in Table III.

3) Indian Pines: The Indian Pines data set is used to illustrate the potential of the proposed method. The data set was captured by the AVIRIS sensor in 1992 over the Indian Pines test site in Northwest Tippecanoe County, IN, USA. The image consists of 145 × 145 pixels with a GSD of 20 m. A total of 16 classes were annotated in the reference data, and most were agricultural land covers. A road exists on the upper side of the image, which should not belong to any of the known classes. We manually annotated the road, as shown in Fig. 6 reference map. Since some of the classes only have less than ten instances, these tail classes were discarded in the experiment following [29] and were considered as unknown, resulting in eight known classes. Details of the data set can be found in Table IV.

TABLE IV
INDIAN PINES

Index	Name	#sample
1	Corn-notill	1428
2	Corn-mintill	830
3	Grass-pasture	483
4	Hay-windrowed	478
5	Soybean-notill	972
6	Soybean-mintill	2454
7	Soybean-clean	593
8	Woods	1265
	Novel	2007
	Total	10510

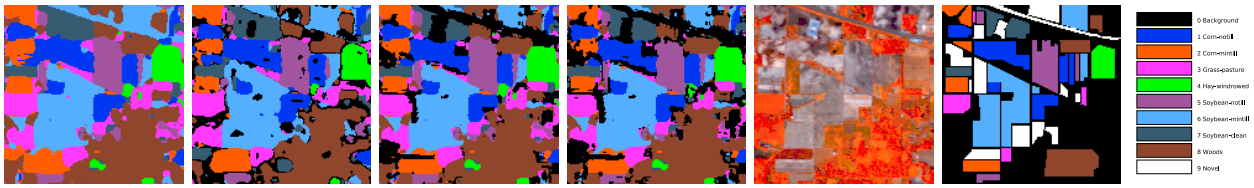
B. Evaluation Metrics

1) Openness: Openness indicates the unknown or open degree of a data set for open-world (OW) classification. It can be calculated from the number of training classes and the number of test classes [68]

$$\text{Openness} = 1 - \sqrt{\frac{2 \times N_{\text{train}}}{N_{\text{test}} + N_{\text{train}}}}. \tag{12}$$

Low openness is one of the differences between daily image recognition and hyperspectral image classification.

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(a) (b) (c) (d) (e) (f)

Fig. 6. Classification maps of the Indian Pines. Known classes: 1–8; unknown classes: 9. In the classification maps, those masked with black color are identified as unknown by the proposed method. (a) Close. (b) CROSR. (c) MDL4OW. (d) MDL4OW/C. (e) Image. (f) Reference.

Some studies pointed out that the performance of certain methods could be degraded with a low openness value [58]. The openness values for the three hyperspectral data sets are 13%, 3%, and 20%, whereas the openness for daily image recognition is up to 63% [67].

2) Open Overall Accuracy (OA): The evaluation of OW classification is critical and diverse because of the inclusion of unknown classes [68]. A naive solution is extending the current OA evaluation metric from C classes to $C+1$ classes, where the added is the unknown class

$$\text{Open OA} = \frac{\sum_{i=1}^{C+1}(\text{TP}_i + \text{TN}_i)}{\sum_{i=1}^{C+1}(\text{TP}_i + \text{TN}_i + \text{FP}_i + \text{FN}_i)} \tag{13}$$

where TP_i, TN_i, FP_i, and FN_i are true positive, true negative, false positive, and false negative of the ith class. In hyperspectral image classification, we are interested in the accuracy of known land cover classes instead of the unknown. However, the open OA metric takes the accuracy of the unknown class into account. If the known classes only account for a small portion of the data, this metric will be dominated by the unknown classes.

The open OA is similar to the closed OA (the one normally used in classification) but includes the unknown class. For reference purposes, we show the closed OA as follows:

$$\text{Closed OA} = \frac{\sum_{i=1}^{C}(\text{TP}_i + \text{TN}_i)}{\sum_{i=1}^{C}(\text{TP}_i + \text{TN}_i + \text{FP}_i + \text{FN}_i)}. \tag{14}$$

3) Micro F1 Score: A more reasonable indicator should be the micro F1 score, where we only consider the precision and recall of known classes. Note the F1 score of the unknown class should not be considered, as pointed by [68]. With micro F1, the inclusion of unknown classes plays an important role in the precision scores. If unknown instances are classified into one of the known classes, precision will be decreased. The micro F1 score is calculated as follows:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

where

$$\text{Precision} = \frac{\sum_{i=1}^{C}\text{TP}_i}{\sum_{i=1}^{C}(\text{TP}_i + \text{FP}_i)} \tag{16}$$

$$\text{Recall} = \frac{\sum_{i=1}^{C}\text{TP}_i}{\sum_{i=1}^{C}(\text{TP}_i + \text{FN}_i)}. \tag{17}$$

4) Mapping Error: The mapping error in a specific region is of direct interest to local governments, as policy-makers rely on food production data collected based on administrative regions to adapt their strategy. In this study, the mapping error is defined as

$$\text{Error} = \frac{\sum_{i=1}^{C}|A_{p,i} - A_{\text{gt},i}|}{\sum_{i=1}^{C}A_{\text{gt},i}} \quad \text{s.t. } A_i \geq 0, \quad \sum_{i=1}^{C+1}A_{p,i} = \sum_{i=1}^{C+1}A_{\text{gt},i} \tag{18}$$

where $A_{p,i}$ is the predicted area of the ith class, $A_{\text{gt},i}$ is the ground truth area, C is the number of known classes, and $C+1$ class is the unknown. Note that, for the closed set classification, the mapping error should be

$$\text{Error}_{\text{close}} = \frac{\sum_{i=1}^{C}|A_{p,i} - A_{\text{gt},i}|}{\sum_{i=1}^{C}A_{\text{gt},i}} \quad \text{s.t. } A_i \geq 0, \quad \sum_{i=1}^{C}A_{p,i} = \sum_{i=1}^{C}A_{\text{gt},i}. \tag{19}$$

If a model overestimates a land cover class, it will also underestimate another land cover class. Note that, in the OW setting, the upper bound of the individual class in theory is $\sum_{i=1}^{C+1}A_{\text{gt},i}$, whereas the upper bound for the closed world assumption is $\sum_{i=1}^{C}A_{\text{gt},i}$. Therefore, the range of mapping error in the closed setting is 0–2, whereas the maximum error in OW classification depends on the unknown instances included for evaluation and is calculated as $\text{Error}_{\max} = 2 \times (1 + A_{\text{gt},C+1}/\sum_{i=1}^{C}A_{\text{gt},i})$.

Mapping error is not a point-by-point evaluation metric, meaning that a zero error does not necessarily lead to zero classification confusion. However, for common multiclass scenarios, it is nearly impossible for a model to have a correct guess on the mapping area randomly. As a well-classified map is supposed to have a good estimation of the land cover area, mapping error is the most simple and direct metric to evaluate the accuracy of hyperspectral image classification, especially with the existence of unknown classes. Mapping error is also sensitive to imbalanced mapping (or in technical terms, the long tail), which is a great challenge when we are facing the open world [69]. We show a toy example in Table V, where three models achieve an identical OA but different errors. A good model, such as Model 1, should carefully take care of the tail classes to reduce the mapping error. In contrast, an imbalanced model, although achieving the same OA, neglects the tail classes and, thus, leads to extreme

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TABLE V

SIMPLE EXAMPLE ILLUSTRATING MAPPING ERROR'S SENSITIVITY TO IMBALANCED MAPPING (THE LONG TAIL). NOTE THAT *a*/*b* INDICATES THE MODEL PREDICTS *b* INSTANCES THAT BELONG TO THE CLASS, WHERE *a* INSTANCES ARE CORRECTLY CLASSIFIED

	C1	C2	C3	OA	Error
GT	80	10	10	-	-
Model 1	70/80	5/10	5/10	80/100	0
Model 2	78/88	1/6	1/6	80/100	16/100
Model 3	80/100	0	0	80/100	40/100

TABLE VI

NUMBER OF KNOWN AND NOVEL CLASSES (IN BRACKETS) AND PIXELS IN THE EXPERIMENTS

		Pavia	Salinas	Indian
Almost perfect	Known Pixels (Classes)	42776 (9)	54129 (16)	8503 (8)
	Novel Pixels (Classes)	5163 (6)	5613 (2)	2007 (9)
OW1	Known Pixels (Classes)	14724 (3)	10716 (3)	2883 (3)
	Novel Pixels (Classes)	33215 (12)	49026 (15)	7627 (14)
OW2	Known Pixels (Classes)	26742 (3)	22483 (3)	4019 (3)
	Novel Pixels (Classes)	21197 (12)	37259 (15)	6491 (14)
OW3	Known Pixels (Classes)	11643 (3)	24069 (3)	3286 (3)
	Novel Pixels (Classes)	36296 (12)	35673 (15)	7224 (14)

overestimation and underestimation of the number and area of each land cover class.

C. Experimental Setup

Experiments were carried out under both the few-shot context (20 training samples per class) and many-shot context (200 per class). The experiments were conducted using TensorFlow 1.9 [70] and Keras 2.1.6 [71] with Python 3.6 on a machine equipped with an Intel I5-8500 CPU, an Nvidia GTX 1080 Ti GPU, and 32-GB RAM. The AdaDelta optimizer [72] is used for backpropagation in the training phase. The learning rate was set as 1.0 for the first 170 epochs and then as 0.1 for another 30 epochs before the training stopped. We adopted the early stopping mechanism to accelerate the training. If the loss does not decrease for five epochs, the learning proceeds immediately to the next phase. To fully exploit the training samples, they were augmented horizontally, vertically, and diagonally; hence, the number of training samples NoS should be times four.

In the experiments, the weights of classification and reconstruction losses are 0.5. The tail number is set as $\text{NoS} \times 4 \times 0.05 \times C$ for MDL4OW and $\text{NoS} \times 4 \times 0.05$ for MDL4OW/C, where NoS is the number of training samples per class, 4 is the number of augmentations of training samples, and C is the number of training classes. If the tail number is smaller than 20 for MDL4OW, then it is set as 20. The tail number here is not the global optimum and will be discussed in Section IV-E. All results are reported from the average of ten random training sets. We compare our method with random forest (RF), SVM, deep contextual CNN (DCCNN) [29], wide contextual residual network (WCRN) [73], the modified HResNet (the baseline of the proposed method) [43], and the state-of-the-art open-set method CROSR [52]. For open-set CNNs with confidence thresholding, the unknown probability is determined with a threshold of 0.5.

IV. RESULTS AND ANALYSIS

The experiments were conducted in two parts. In the first part, we evaluated the proposed method in an almost perfect classification system. Under this setting, only a small portion of the instances belongs to unknown or novel due to the long-tail nature of the open world. In the second part, we selected some known classes as training data and the other classes as the unknown to evaluate the proposed method. This is the standard protocol for the open classification of daily images in computer vision, and the number of unknown classes greatly exceeds the known classes. However, the second setting is rare in real-world hyperspectral image classification. The number of known and novel classes and their supporting pixels are shown in Table VI.

A. With Almost Perfect Classification Systems

1) Results on the University of Pavia Data Set: In the first experiment, we applied the proposed method to the popular benchmark data from the University of Pavia. In Tables VII and VIII, we show the overall OA, F1, mapping error, and classwise F1 obtained with 20 and 200 training samples per class. Under the few-shot setting, MDL4OW achieved the best classification in terms of three evaluation metrics (85.07%, 0.9172, 14.03%), whereas CROSR achieved the second-best F1 score (84.66%, 0.9090, 14.95%). Under the many-shot setting, the best classification was obtained by MDL4OW/C. The OA and F1 are 90.53% and 0.9482, 1.87% and 0.0083 higher than the closed classification. When it comes to the mapping error, the proposed method significantly outperformed conventional methods. The mapping error was reduced by 4.9% from 12.19% to 7.29%.

For illustrative purposes, we show the classification maps in Fig. 4. We can see that the proposed method successfully identified the unknown classes (not annotated in the standard data set) building-1, vehicles (the helicopter and some cars), the parking shed, and the swimming pool. Among the three open methods, only the CROSR identified building-2 as unknown, which is reflected in its high F1 score on the novel class. However, CROSR achieved a lower F1 in the second class meadows. This class covers a large area in the image, as indicated by the number of sample instances (see Table II). As shown in the classification map [see Fig. 4 (CROSR)], on the lower side of the image, some meadows were misclassified as unknown by CROSR. CROSR is a centroid-based method and is sensitive to the intraclass variation. Meadows consist of over 18 000 instances and have a large intraclass variation, limiting the performance of centroid-based methods. Large intraclass variation is common in remote sensing images. Therefore, the proposed MDL4OW is more robust and stable compared with CROSR.

2) Results on the Salinas Valley Data Set: To evaluate the potential of the proposed method in estimating crop area, a popular agricultural data set, the Salinas Valley, is used in the experiment. This data set contains 16 agricultural classes, where some of the man-made materials are not considered

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TABLE VII

CLASSIFICATION ACCURACY USING 20 SAMPLES PER CLASS ON THE UNIVERSITY OF PAVIA DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	76.34	75.82	84.60	85.50	91.66	84.60	85.52	91.72	85.87	91.56	89.78
2	69.17	81.67	91.15	93.77	95.57	91.17	93.76	95.57	93.37	95.57	93.84
3	51.82	60.32	57.50	67.46	79.82	57.78	67.71	80.15	83.41	82.70	81.95
4	68.44	78.96	89.98	91.09	92.16	90.08	91.23	92.35	92.96	91.69	89.47
5	64.38	67.13	84.71	89.02	88.46	84.92	89.11	88.78	95.83	90.27	91.88
6	40.06	53.04	65.76	67.50	74.81	65.81	67.68	75.00	77.91	76.37	78.32
7	61.01	62.79	64.78	72.19	72.24	65.03	72.59	72.94	90.57	75.71	76.01
8	67.41	75.17	72.24	70.35	82.53	72.41	70.50	82.62	84.38	83.93	80.42
9	99.79	99.88	91.27	91.23	86.63	91.51	91.46	87.25	93.19	86.72	84.51
Novel	0.00	0.00	0.00	0.00	0.00	2.50	2.73	4.87	57.21	32.77	42.06
OA (%)	60.36±2.19	69.28±3.22	76.81±2.78	79.40±1.56	83.52±1.42	76.90±2.82	79.44±1.56	83.73±1.36	84.66±3.61	**85.07±1.58**	83.35±2.43
F1×100	75.25±1.71	81.81±2.30	86.86±1.81	88.51±0.97	91.01±0.84	86.89±1.83	88.51±0.97	91.11±0.81	90.90±2.28	**91.72±0.96**	90.44±1.50
Error (%)	44.22±9.17	30.27±6.26	24.05±6.67	17.19±4.40	16.59±3.31	23.81±6.72	16.93±4.51	16.31±3.22	14.95±5.23	**14.03±3.69**	14.31±3.69

TABLE VIII

CLASSIFICATION ACCURACY USING 200 SAMPLES PER CLASS ON THE UNIVERSITY OF PAVIA DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	82.77	83.22	94.90	93.44	95.29	94.88	93.46	95.34	93.05	93.68	94.54
2	83.73	92.93	97.84	98.56	99.03	97.85	98.56	99.04	96.64	99.03	98.81
3	72.67	76.75	78.36	93.00	93.75	78.54	93.03	93.82	93.96	97.39	96.72
4	83.37	92.24	95.83	97.01	90.16	95.89	97.04	90.24	95.02	90.42	90.61
5	75.79	75.08	90.53	94.61	88.74	90.61	94.64	88.86	96.08	98.11	97.47
6	52.88	73.54	76.59	77.48	83.93	76.66	77.50	84.01	86.55	85.23	86.29
7	72.67	79.91	84.09	75.77	75.59	84.35	75.83	75.77	96.17	81.93	80.56
8	80.28	83.63	88.75	87.48	96.65	88.94	87.53	96.71	95.04	94.81	95.74
9	99.51	99.92	97.31	96.04	94.46	97.39	96.07	94.61	96.29	95.21	94.52
Novel	0.00	0.00	0.00	0.00	0.00	1.85	0.52	1.91	61.70	38.43	45.94
OA (%)	72.77±0.88	81.36±0.43	86.55±0.50	87.39±0.18	88.66±0.12	86.58±0.54	87.41±0.17	88.76±0.13	90.39±0.67	90.17±0.13	**90.53±0.62**
F1×100	84.24±0.59	89.72±0.26	92.78±0.29	93.27±0.10	93.99±0.07	92.80±0.31	93.28±0.10	94.03±0.07	94.51±0.37	94.66±0.08	**94.82±0.33**
Error (%)	29.87±2.34	14.72±1.62	13.76±1.14	12.66±0.48	12.19±0.18	13.56±1.15	12.61±0.49	12.06±0.18	7.39±1.44	8.36±0.04	**7.29±0.95**

TABLE IX

CLASSIFICATION ACCURACY USING 20 SAMPLES PER CLASS ON THE SALINAS VALLEYS DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	97.25	97.64	80.93	92.02	99.86	81.02	92.09	99.86	93.24	94.58	94.49
2	97.80	97.90	85.30	98.05	99.74	85.21	98.07	99.77	92.60	96.95	97.99
3	80.22	91.99	73.05	89.78	97.90	72.42	89.74	97.93	93.27	98.58	95.42
4	70.62	75.44	71.92	74.86	69.31	72.03	74.99	69.42	96.23	97.22	95.93
5	90.24	92.25	77.16	71.11	67.05	77.36	71.26	67.13	93.86	83.50	85.75
6	81.66	90.72	95.75	93.97	97.49	95.83	94.07	97.60	93.17	95.19	99.12
7	96.87	98.49	98.09	98.38	98.59	98.09	98.37	98.55	94.67	96.31	95.59
8	62.70	70.79	39.08	71.62	74.71	35.33	71.44	74.72	74.66	75.22	75.18
9	93.38	87.11	89.36	94.57	97.24	89.48	94.57	97.26	95.12	98.02	96.81
10	74.87	77.92	81.30	84.25	90.97	81.57	84.32	91.04	89.11	91.89	90.28
11	62.30	61.27	63.14	78.83	73.03	63.61	79.50	73.35	84.83	82.37	81.84
12	92.82	94.40	91.20	98.51	99.54	91.93	98.52	99.55	93.61	99.89	98.70
13	89.68	93.88	90.60	97.28	99.16	91.83	97.29	99.22	94.80	99.93	99.56
14	80.83	88.93	92.94	94.30	98.54	93.20	94.42	98.58	95.37	98.83	96.65
15	56.67	62.46	54.08	65.87	77.40	49.56	65.96	77.40	76.45	77.49	77.31
16	84.67	87.57	93.90	80.50	90.89	94.11	80.67	90.94	94.64	99.14	95.43
Novel	0.00	0.00	0.00	0.00	0.00	4.21	2.32	1.69	64.94	74.39	75.97
OA (%)	74.50±1.63	78.02±1.76	70.48±3.19	78.74±1.69	82.46±1.65	68.84±5.69	78.69±1.77	82.51±1.64	84.27±1.29	**87.40±1.60**	87.12±1.57
F1×100	85.38±1.08	87.64±1.11	82.64±2.17	88.09±1.06	90.38±1.39	81.35±4.19	88.05±1.11	90.41±1.41	90.57±0.85	**92.73±1.36**	92.43±1.37
Error (%)	19.91±5.19	17.37±4.50	36.71±14.27	21.12±6.09	23.24±5.59	36.62±13.66	20.97±6.14	23.15±5.59	18.68±2.12	**17.04±5.37**	18.20±5.67

(a road on the lower left and a water pool on the upper left). We manually annotated some of the man-made materials in white, as shown in Fig. 5 (reference map). The classification results are presented in Tables IX and X for few- and many-shot settings. Under the few-shot context, MDL4OW achieved the best classification result in terms of three evaluation metrics. Compared with the close classification, the OA increased by 4.94% from 82.46% to 87.40%; the F1 score increased by 0.0235 from 0.9038 to 0.9273. The mapping error is more sensitive to the unknown classes and reduced from 23.24% to 17.04%, up to 6%. This experiment strongly shows the potential of the proposed method in precisely estimating crop area. As for the many-shot setting, the classwise MDL4OW/C achieved the best classification results, followed by MDL4OW. The mapping error was significantly reduced from 10.84% to 3.96%.

For illustrative purposes, we show the classification maps obtained from the many-shot setting in Fig. 5. We can see that a lot of known instances (the lower part) are misclassified as unknown with the CROSR method. In contrast, the proposed MDL4OW successfully identified the unknown (a road on the lower left and a water pool on the upper left) while maintaining high accuracy on the known classes. As in remotely sensed hyperspectral images, the unknown should only count for a small portion of the image; the proposed method is more suitable for the classification of hyperspectral data.

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TABLE X

CLASSIFICATION ACCURACY USING 200 SAMPLES PER CLASS ON THE SALINAS VALLEYS DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	99.66	99.83	84.00	87.70	91.18	84.10	87.81	91.30	96.58	94.01	94.60
2	99.63	99.73	99.35	99.99	99.93	99.40	99.99	99.94	97.05	97.74	99.04
3	91.53	97.53	93.92	88.70	88.94	94.18	88.79	89.02	96.25	92.18	98.45
4	71.94	81.47	88.35	80.35	64.48	88.37	80.45	64.53	97.12	98.19	98.55
5	92.54	85.04	73.29	74.36	87.25	73.37	74.46	87.36	96.97	95.17	96.47
6	85.91	93.21	98.14	97.94	98.80	98.19	98.00	98.83	96.82	99.73	97.26
7	98.56	99.34	99.51	99.68	95.57	99.53	99.68	95.60	96.86	97.16	97.87
8	75.99	82.22	80.59	85.64	90.95	80.54	85.65	90.95	90.17	91.71	92.55
9	96.76	94.76	94.99	96.03	95.99	95.01	96.05	96.01	96.36	97.44	97.72
10	74.97	74.49	89.85	89.02	93.51	90.09	89.18	93.61	94.35	96.24	97.39
11	73.20	91.95	84.13	99.05	96.55	84.42	99.08	96.61	96.35	97.48	97.53
12	97.25	98.92	99.69	99.91	99.79	99.71	99.91	99.80	96.76	99.85	97.76
13	97.16	98.85	99.68	99.75	99.56	99.71	99.76	99.61	97.15	99.91	99.12
14	91.60	94.26	94.74	99.19	97.70	95.03	99.21	97.81	96.12	98.79	98.51
15	67.87	75.37	79.08	82.16	91.07	79.13	82.21	91.08	88.70	91.18	89.99
16	93.92	94.59	98.40	93.94	92.84	98.47	93.97	92.86	96.83	99.61	98.62
Novel	0.00	0.00	0.00	0.00	0.00	2.84	1.95	1.57	75.26	76.87	83.32
OA (%)	80.85±0.50	83.87±0.30	84.08±2.19	85.72±1.60	88.14±1.90	84.16±2.16	85.81±1.60	88.22±1.90	91.82±0.52	93.73±0.80	**94.34±0.61**
F1×100	89.41±0.30	91.23±0.17	91.34±1.32	92.30±0.95	93.70±1.50	91.37±1.31	92.35±0.95	93.73±1.51	95.29±0.31	96.53±0.60	**96.82±0.71**
Error (%)	15.16±2.08	11.53±0.90	17.31±7.29	14.30±5.24	10.84±4.50	17.12±7.30	14.21±5.24	10.76±4.50	6.98±0.83	4.63±2.79	**3.96±2.61**

TABLE XI

CLASSIFICATION ACCURACY USING 20 SAMPLES PER CLASS ON THE INDIAN PINES DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	47.95	48.43	50.24	62.79	73.48	50.23	62.92	73.54	71.58	74.89	74.51
2	46.15	41.52	39.08	51.76	69.12	39.42	52.22	69.35	68.63	68.22	68.45
3	35.75	44.91	62.45	62.52	67.03	62.70	63.48	67.55	70.05	65.12	66.34
4	82.43	89.64	90.58	90.24	89.57	90.90	90.88	89.79	88.04	89.61	89.24
5	53.92	54.54	46.07	60.47	76.42	45.93	61.08	76.58	75.03	76.79	78.21
6	54.45	54.87	56.35	64.98	77.06	56.28	65.06	77.01	74.43	77.23	76.55
7	37.06	45.99	43.56	48.42	65.57	43.79	48.92	66.10	66.99	65.50	66.01
8	87.23	83.34	75.06	84.99	84.49	75.40	85.52	84.64	87.85	84.34	83.85
Novel	0.00	0.00	0.00	0.00	0.00	6.98	9.70	3.11	36.16	21.18	27.93
OA (%)	49.27±1.48	51.01±1.90	50.99±2.98	58.67±2.10	67.98 ±2.58	51.35±2.87	59.35±2.35	68.14±2.57	68.49±3.94	68.92±3.26	**69.00±3.90**
F1×100	66.00±1.33	67.53±1.68	67.49±2.61	73.93±1.68	80.91±1.87	67.43±2.67	74.12±1.81	80.95±1.87	79.83±2.40	**80.99±2.19**	80.73±2.42
Error (%)	46.48±8.97	41.74±9.19	48.43±9.27	38.26±8.64	33.27±10.10	47.11±11.16	36.71±9.62	32.62±9.93	28.90±11.85	**28.32±10.74**	28.37±11.30

TABLE XII

CLASSIFICATION ACCURACY USING 200 SAMPLES PER CLASS ON THE INDIAN PINES DATA SET. * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN. ERROR (%): THE SMALLER THE BETTER; ELSE: THE LARGER THE BETTER

	Close					Open					
	RF	SVM	DCCNN [29]	WCRN [73]	HResNet [43]	DCCNN*	WCRN*	HResNet*	CROSR [52]	MDL4OW	MDL4OW/C
1	76.23	78.72	77.63	94.14	94.97	77.72	94.22	95.06	94.34	95.31	95.03
2	72.88	72.83	73.07	84.24	82.66	72.52	84.40	82.77	82.89	83.26	83.47
3	54.32	54.74	64.01	71.76	85.79	64.51	72.31	86.31	86.05	85.33	85.66
4	91.77	93.89	95.12	95.01	95.17	95.15	95.05	95.23	95.91	94.88	95.31
5	78.01	77.04	81.76	91.28	93.79	82.16	91.65	94.08	93.97	94.08	94.52
6	77.84	78.80	82.44	94.80	87.64	81.57	94.90	87.80	87.12	88.82	88.67
7	55.72	85.09	81.23	85.49	82.91	81.86	85.95	83.44	88.76	84.98	88.35
8	79.32	86.43	75.90	75.73	86.91	76.03	76.04	87.16	88.91	87.58	87.70
Novel	0.00	0.00	0.00	0.00	0.00	3.86	4.76	4.50	22.09	32.21	33.58
OA (%)	66.73±0.58	70.53±0.53	70.97±6.56	79.13±0.54	80.14±0.40	70.41±7.29	79.58±0.56	80.57±0.46	81.55±1.53	82.23±1.64	**82.61±1.91**
F1×100	80.04±0.04	82.72±0.36	82.84±4.62	88.35±0.34	88.97±0.24	82.32±5.24	88.57±0.35	89.19±0.27	89.54±0.79	89.79±0.89	**90.01±1.02**
Error (%)	34.95±1.70	28.13±4.02	33.75±8.95	24.54±1.81	23.60±0.23	33.22±9.02	23.95±1.81	23.00±0.31	18.76±2.17	16.26±1.49	**16.00±2.36**

3) Results on the Indian Pines Data Set: To show the generalization of the proposed method, another popular data set, Indian Pines, is tested. This data set, as shown in the false-color map in Fig. 6, mainly covers an agricultural region. In the standard 16-class reference map, one nonagricultural class exists. However, the road on the upper side of the image is not represented in the reference data. It should not be considered as an agricultural class. As mentioned earlier, since some of the land covers in this data set consist of very few samples, we only included eight classes as the training set following [29]. The remaining $8+1$ classes were naturally considered as unknown. The results are reported in Tables XI and XII.

Under the few-shot context, the three open methods are competitive and resulted in a mapping error of 28%–29%, 3%–4% lower compared with the closed error of 33.27%, whereas the proposed MDL4OW is slightly better. As for the many-shot setting, the classwise MDL4OW/C obtained the best classification result in three evaluation metrics. The OA and F1 are 82.61% and 0.9001, 2.47% and 0.0104 higher than the closed classification. The mapping error is more sensitive and is 16.00%, 7.60% lower than the baseline.

We show the classification maps obtained using 200 samples per class in Fig. 6. The road on the upper side of the image is successfully identified as unknown by the proposed method, both MDL4OW and MDL4OW/C. The close CNN classified

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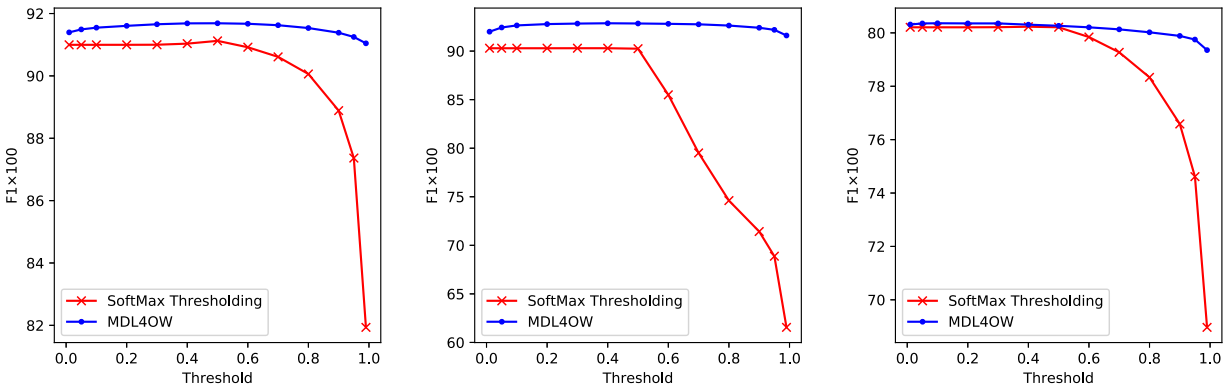


Fig. 7. Confidence threshold analysis in terms of F1 score under the few-shot setting. (a) Pavia. (b) Salinas. (c) Indian.

TABLE XIII

COMPARISON WITH OTHER METHODS IN TERMS OF CLOSED OA (%) AND OPEN OA (%). * TOP-1 SOFTMAX PROBABILITY < 0.5 IS CONSIDERED AS UNKNOWN

Method		Pavia University		Salinas Valley	
		Closed OA	Open OA	Closed OA	Open OA
Non-deep	RF	81.56±0.99	72.77±0.88	89.24±0.55	80.85±0.50
	SVM	91.18±0.48	81.36±0.43	92.60±0.34	83.87±0.30
Close	DCCNN [29]	96.99±0.56	86.55±0.50	92.84±2.41	84.08±2.19
	WCRN [73]	97.94±0.20	87.39±0.18	94.66±1.76	85.72±1.60
	HResNet	99.36±0.15	88.66±0.12	97.39±0.64	88.14±1.90
	CNN [74]	87.90	78.43	90.25	81.77
	R-PCA-CNN [42]	94.38	84.62	92.39	83.71
	CNN-PPF [75]	96.48	86.09	94.80	85.89
	C-CNN [76]	98.41	87.81	97.42	88.27
	SSRN [35]	99.79	89.04	98.68	89.41
Few-shot	DFSL+NN [36]	97.85	87.31	97.78	88.59
	DFSL+SVM [36]	98.62	88.00	97.81	88.62
	SS-RN [37]	99.69	88.95	98.62	89.35
Open	DCCNN*	96.92±0.59	86.58±0.54	92.77±2.41	84.16±2.16
	WCRN*	97.93±0.20	87.41±0.17	94.65±1.76	85.81±1.60
	HResNet*	99.35±0.14	88.76±0.13	97.38±0.64	88.22±1.90
	CROSR [52]	96.67±0.25	90.39±0.67	94.75±0.53	91.82±0.52
	MDL4OW	97.93±0.26	90.17±0.13	96.37±0.80	93.73±0.80
	MDL4OW/C	97.23±0.37	90.53±0.62	94.99±0.63	94.34±0.61

it as one of the agricultural classes, soybean-mintill, leading to its overestimation. CROSR failed to recognize the road as unknown. From the classification maps, we can see that most man-made materials unrelated to crops can be identified as unknown by the proposed method, showing its promise in precise crop mapping.

4) Threshold Analysis: We show a threshold analysis under the few-shot context in Fig. 7 to demonstrate that: 1) SoftMax confidence thresholding is not enough for detecting the unknown and 2) MDL4OW is robust to the threshold setting (z). From Fig. 7, we can see that with a larger threshold, the F1 score first slightly improves and then drops sharply. A low SoftMax probability indicates that the classifier is uncertain with its decision, but uncertainty is not equal to the unknown. A sample instance that is difficult to classify may be a valuable instance close to the decision boundary between two classes (the valuable sample in active learning) [46].

B. Comparison With Other Methods

In Table XIII, we compare the proposed methods with other competitors in terms of closed OA and open OA using 200 samples per class. The comparison is based on Pavia and Salinas data. We do not compare the results on the Indian data since the used known classes are different from other studies. Closed OA is calculated based on known classes without considering the existence of an unknown class. Open OA includes the unknown class. It is easy to convert a closed OA to open OA once the unknown instances are determined from a reference map. Besides two nondeep methods (RF and SVM), we also include eight deep learning methods and three deep learning-based few-shot methods. They are DCCNN [29], WCRN [73], HResNet (baseline of the proposed method), CNN [74], R-principal component analysis (PCA)-CNN [42], CNN-pixel-pair features (PPF) [75], C-CNN [76], spectral-spatial residual network (SSRN) [35], DFSL + NN [36], DFSL + SVM [36], and SS-RN [37].

MDL4OW/C achieves the best open OA, followed by MDL4OW. Those closed methods lack the ability to tell the unknown from known classes and, therefore, achieve a low open OA. Take the SSRN [35] as an example. It achieves the highest closed OA on the Salinas data (98.68%) but only obtains an open OA of 89.41%. MDL4OW/C obtains the highest open OA (94.34%), whereas MDL4OW obtains the second-highest open OA (93.73%) and a reasonable closed OA (96.37%). It should note that although open methods are promising in identifying the novel class, they inevitably lead to either a large or a slight decrease in the closed OA since the closed methods do not bother with the novel class. The tradeoff here is similar to target detection [77]. As the novel class is inevitable in the real world, a method that is capable of identifying the unknown and maintaining good performance on the known classes is essential. The proposed MDL4OW achieves a good tradeoff here if we look at both the closed OA and open OA.

C. With Imperfect Classification Systems

In this section, we evaluate the proposed method with imperfect classification systems. Land cover classification systems are multilevel. For fine-grained crop mapping, each crop can be categorized as an individual class, but they all belong to a larger class, vegetation. The hierarchy of land cover classification systems makes it easy to distinguish between impervious and vegetation but difficult to distinguish between meadows and trees. Therefore, we designed a series

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TABLE XIV

SELECTED IMPERFECT CLASSIFICATION SYSTEMS. THE NUMBER IS THE CLASS INDEX. * INCOMPLETE CLASSIFICATION SYSTEM

Dataset	#Train	#Test	Openness (%)	OW1	OW2	OW3
Pavia	3	15	42.3	[1, 4, 6] Asphalt, Trees, Bare S.	[2, 4, 6]* Meadows, Trees, Bare S.	[1, 7, 8]* Asphalt, Bitumen, Brick
Salinas	6	18	29.3	[1, 2, 11, 12, 13, 14]* Weed-1, Weed-2, Lett.-4wk Lett.-5wk, Lett.-6wk, Lett.-7wk	[1, 3, 6, 9, 11, 15] Weed-1, Fallow, Stubble Soil, Lett.-4wk, Viny.-U	[2, 5, 9, 10, 13, 15] Weed-2, Fallow-S, Soil Corn, Lett.-6wk, Viny.-U
Indian	3	17	45.2	[1, 3, 5] Corn-n., Grass, Soybean-n.	[5, 6, 7]* Soybean-n., Soybean-m., Soybean-c.	[1, 7, 8] Corn-n., Soybean-c., Woods

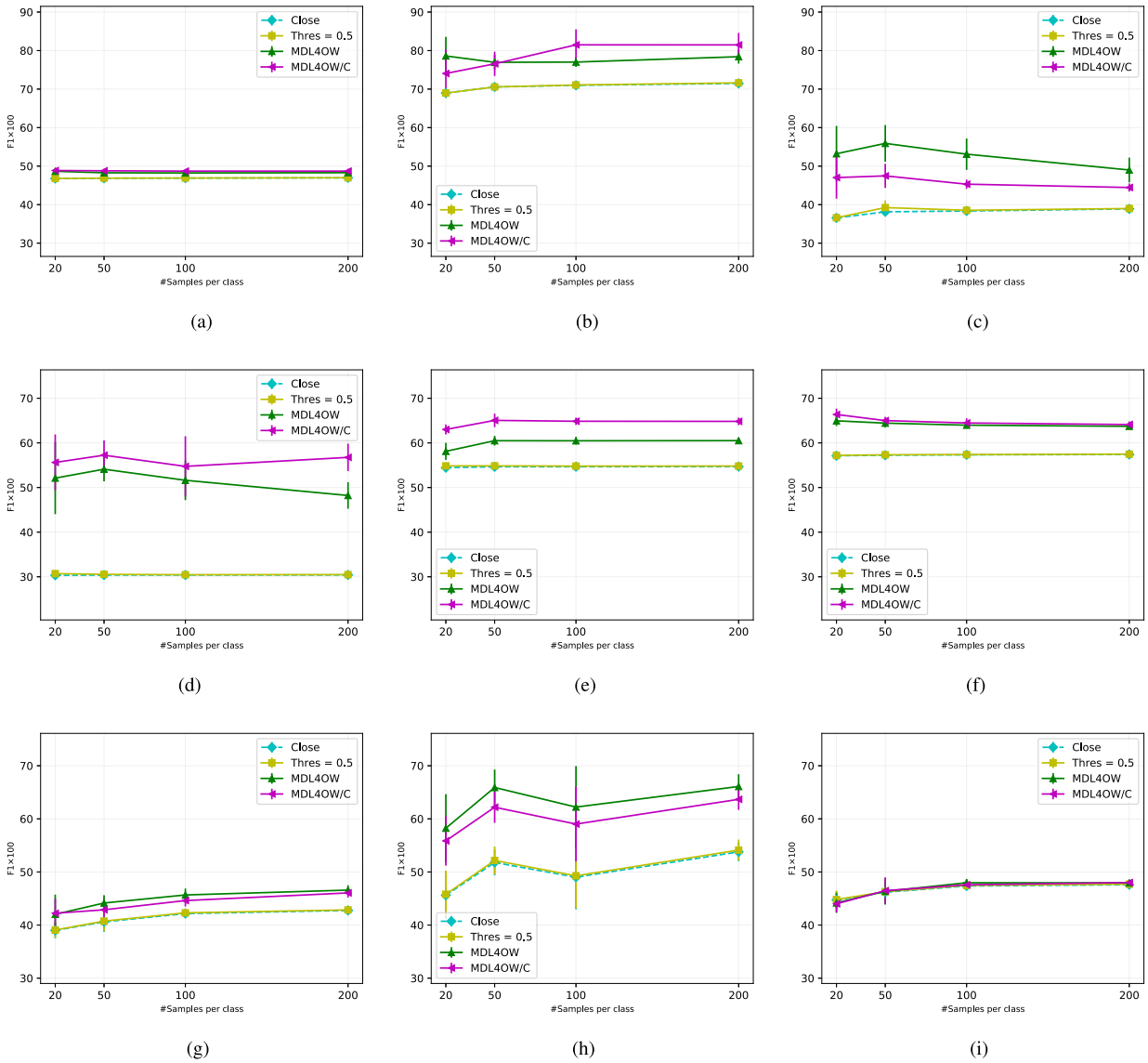


Fig. 8. F1 as a function of the number of training samples per class. * Incomplete classification system. (a) Pavia, OW1. (b) Pavia, OW2*. (c) Pavia, OW3*. (d) Salinas, OW1*. (e) Salinas, OW2. (f) Salinas, OW3. (g) Indian, OW1. (h) Indian, OW2*. (i) Indian, OW3.

of experiments, where we selected some classes into training, as shown in Table XIV. Some of them are all from impervious classes, leading to a classification system without vegetation (Pavia OW3); some of them are the same crop type with a minor difference in terms of growing weeks (Salinas OW1).

A classification system lacking the basic elements (impervious, vegetation, and water) is extremely incomplete and in need of the proposed method to reject the unknown classes. For a complete classification system, this task will be extremely difficult, as the interclass variation might be too small. For the experimental details, we select three, six, and three training classes for the Pavia, Salinas, and Indian data sets, respectively, and all classes as the test set, as shown in Table XIV, leading to openness values of 42.3%, 29.3%, and 45.2%, respectively. Three OW scenarios are tested for

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TABLE XV

CLASSIFICATION COMPARISON ON THE PAVIA DATA SET: OA (%), F1 × 100, AND MAPPING ERROR (%). OPENNESS = 42.3%. * INCOMPLETE CLASSIFICATION SYSTEM. DUE TO THE EXISTENCE OF UNKNOWN INSTANCES, THE MAXIMUM ERROR CAN BE LARGER THAN 100% AND IS CALCULATED AS $\text{ERROR}_{\max} = 2 \times (1 + A_{\text{gt},C+1} / \sum_{i=1}^{C} A_{\text{gt},i})$

	Method	Few-shot			Many-shot		
		OA	F1	Error	OA	F1	Error
OW1	Close	30.53	46.78	225.58	30.68	46.96	225.58
	Softmax	30.59	46.80	225.38	30.75	46.98	225.35
	CROSR	38.94	25.53	196.67	38.17	25.32	201.25
	MDL4OW	**39.94**	48.66	**187.63**	**38.18**	48.29	**189.38**
	MDL4OW/C	38.85	**48.86**	189.62	36.96	**48.71**	200.18
OW2*	Close	52.65	68.98	82.33	55.61	71.47	79.29
	Softmax	52.65	68.96	82.19	55.94	71.62	78.69
	CROSR	33.67	29.56	114.32	28.12	22.85	116.41
	MDL4OW	**71.71**	**78.60**	**44.03**	70.21	78.41	48.26
	MDL4OW/C	63.99	74.06	58.60	**75.00**	**81.48**	**41.98**
OW3*	Close	22.37	36.56	311.74	24.15	38.90	311.74
	Softmax	22.62	36.61	310.40	24.46	38.99	310.43
	CROSR	25.19	19.26	307.32	30.23	18.78	287.55
	MDL4OW	**61.44**	**53.22**	**141.99**	**52.54**	**49.00**	**182.28**
	MDL4OW/C	50.19	47.02	193.03	41.04	44.45	237.39

each data set: some are complete, and some are incomplete (denoted with *).

The F1 score as a function of the number of training samples per class is shown in Fig. 8. As shown, the performance of the proposed method varies. For the incomplete classification systems (with *), the proposed MDL4OW significantly improves the classification in terms of the F1 score. For the others, the improvement is less significant. Interestingly, the accuracy in terms of F1 is not affected by the number of training samples per class. In this case, collecting more samples cannot enhance the classification in the OW with the existence of unknown classes. This provides the insight that, for hyperspectral image classification, a machine learning model needs to identify the unknown classes; otherwise, the classification accuracy cannot be improved by simply adding more data, as those unknowns will decrease the precision because the model is forced to assign a label.

More details compared with the state-of-the-art methods in terms of the three evaluation metrics (OA, F1, and mapping error) are shown in Tables XV–XVII for the Pavia, Salinas, and Indian data sets, respectively. The best classification is highlighted in bold in each setting. In most cases, the proposed method outperforms competitors.

Although, in theory, MDL4OW is supposed to have a better classification compared with MDL4OW/C under the few-shot context, in this experiment, MDL4OW/C can sometimes achieve very good results. The problem of the classwise method is that, when it comes to few-shot learning, there are not enough instances to estimate the distribution. However, if the openness is large, meaning that the unknown data exceed the known data (this is rare in hyperspectral image classification since it is not difficult to define an acceptable classification system where there are more knowns than unknowns), then the overlapped region of the known and unknown instances would be large enough. In this case, the tail size of the known instances is large enough; even a small set of training data can achieve a reasonable estimation of the distribution. More details about the tail analysis can be found in Section IV-E.

D. Reconstruction Analysis

Since a critical assumption of the proposed method is that the unknown classes will be poorly reconstructed, we show the original and reconstructed spectral profiles of the Pavia data set in Fig. 9. Trees are one of the known classes in the Pavia data set. The reconstructed spectral profiles remain as the classic vegetation curve. The reconstructed profiles are tighter, indicating less intraclass variation. Building-1 is one of the unknown classes. We can see that the reconstructed spectral profiles are very different from the original ones. This analysis confirms the assumption that the unknown classes are poorly reconstructed, indicating the effectiveness of the proposed method.

TABLE XVI

CLASSIFICATION COMPARISON ON THE SALINAS DATA SET: OA (%), F1 × 100, AND MAPPING ERROR (%). OPENNESS = 29.3%. * INCOMPLETE CLASSIFICATION SYSTEM. DUE TO THE EXISTENCE OF UNKNOWN INSTANCES, THE MAXIMUM ERROR CAN BE LARGER THAN 100% AND IS CALCULATED AS $\text{ERROR}_{\max} = 2 \times (1 + A_{\text{gt},C+1} / \sum_{i=1}^{C} A_{\text{gt},i})$

	Method	Few-shot			Many-shot		
		OA	F1	Error	OA	F1	Error
OW1*	Close	17.87	30.33	457.54	17.93	30.41	457.50
	Softmax	19.34	30.71	449.38	18.21	30.48	455.98
	CROSR	**73.24**	17.91	**144.16**	51.82	7.17	257.43
	MDL4OW	66.96	52.11	178.38	63.09	48.22	199.96
	MDL4OW/C	71.80	**55.63**	151.29	**73.16**	**56.75**	**145.44**
OW2	Close	37.40	54.44	165.72	37.63	54.68	165.72
	Softmax	38.42	54.84	162.94	37.96	54.81	164.85
	CROSR	45.09	31.09	144.81	42.29	29.66	153.30
	MDL4OW	47.41	58.11	135.71	52.16	60.50	124.12
	MDL4OW/C	**57.75**	**63.00**	**106.92**	**60.05**	**64.82**	**102.06**
OW3	Close	39.97	57.12	148.21	40.26	57.41	148.21
	Softmax	40.16	57.19	147.71	40.43	57.47	147.79
	CROSR	48.92	32.17	126.45	50.71	33.03	122.28
	MDL4OW	58.12	64.95	96.97	55.39	63.72	105.17
	MDL4OW/C	**61.71**	**66.37**	**83.93**	**55.96**	**64.11**	**105.01**

TABLE XVII

CLASSIFICATION COMPARISON ON THE INDIAN DATA SET: OA (%), F1 × 100, AND MAPPING ERROR (%). OPENNESS = 45.2%. * INCOMPLETE CLASSIFICATION SYSTEM. DUE TO THE EXISTENCE OF UNKNOWN INSTANCES, THE MAXIMUM ERROR CAN BE LARGER THAN 100% AND IS CALCULATED AS $\text{ERROR}_{\max} = 2 \times (1 + A_{\text{gt},C+1} / \sum_{i=1}^{C} A_{\text{gt},i})$

	Method	Few-shot			Many-shot		
		OA	F1	Error	OA	F1	Error
OW1	Close	24.24	39.00	264.55	27.19	42.76	264.55
	Softmax	24.50	39.07	263.34	27.49	42.85	263.46
	CROSR	36.25	41.54	206.9	31.96	44.31	245.84
	MDL4OW	**36.99**	42.02	**205.39**	**41.78**	**46.58**	**198.04**
	MDL4OW/C	36.92	**42.22**	207.55	37.77	46.05	221.29
OW2*	Close	29.65	45.61	171.87	36.80	53.78	161.84
	Softmax	30.33	45.80	169.56	37.65	54.12	159.58
	CROSR	34.77	18.27	144.18	42.55	18.61	109.71
	MDL4OW	**60.08**	**58.24**	**87.09**	**64.31**	**66.10**	**79.19**
	MDL4OW/C	54.84	55.86	104.96	58.89	63.68	100.12
OW3	Close	28.82	44.73	219.84	31.23	47.60	219.84
	Softmax	29.04	44.80	219.09	31.54	47.71	218.86
	CROSR	**42.13**	**48.29**	**160.43**	**49.63**	45.11	**123.94**
	MDL4OW	33.08	46.18	190.60	35.97	47.95	193.51
	MDL4OW/C	31.06	46.00	201.28	34.02	**48.00**	205.96

E. Sensitivity of the Tail Number

The tail number τ is an important factor in the proposed method and should be set carefully. We show the OA, F1, and mapping error as a function of the tail number under the few-shot context in Fig. 10. We can see that the proposed MDL4OW is less sensitive to the tail number than MDL4OW/C. The former is conducted on all training data and free from estimating the class centroid in a classwise fashion. Under the few-shot context, since the number of training samples per class is limited (20 per class), it is difficult to apply the EVT to estimate the extreme cases. Note that we augmented the training samples so the number of training samples per class should be 4×, which, in a way, eases the problem.

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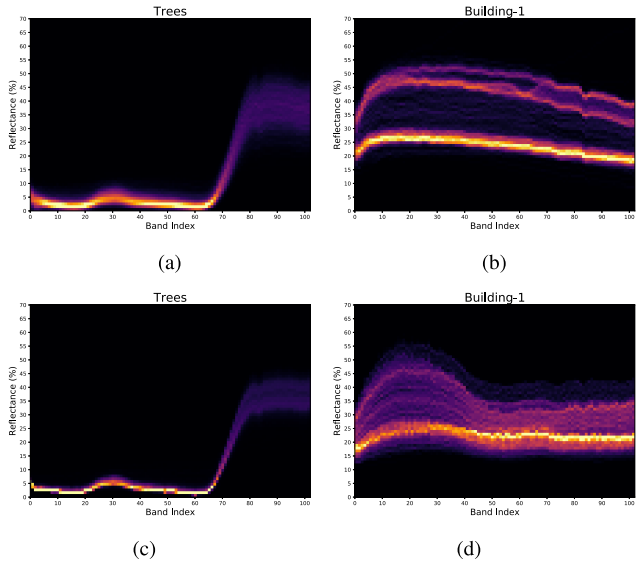


Fig. 9. Density of the original (top) and reconstructed (bottom) spectral profiles of the University of Pavia data set. A brighter color indicates that more sample instances have the similar spectral profiles. (a) Original trees. (b) Original building-1. (c) Reconstructed trees. (d) Reconstructed building-1.

For the Pavia and Indian data sets, the F1 scores are already lower than the closed classification when the tail size τ is 2 (EVT does not work when the tail size τ is 1 because one instance cannot estimate a distribution). For the Salinas data set, the safe range of the tail number in terms of F1 is 2–12, where $\tau = 2$ is the optimum, showing that a smaller tail should be set, but there are not enough instances in the classwise method.

Comparing three evaluation metrics, F1 is always higher than OA and is stricter in evaluation. Take Salinas MDL4OW/C as an example. F1 is lower than the baseline when the tail number is greater than 12, but OA is still higher than the baseline until the tail number is greater than 17. Recall that the F1 score is calculated based on the known classes, and OA includes the unknown class. When the tail size is between 12 and 17, the accuracy of the unknown class makes OA above the baseline. However, since we care more about the known classes than the unknown, it makes more sense to use the F1 score instead of OA. As for mapping error, its real meaning is the average error on the predicted number of all known classes (i.e., land cover area) and is of direct interest to local governments. We can see that it benefits the most from the proposed method where the unknown classes are considered. The evaluation of open-set classification is still an ongoing problem [68]. Although OA and F1 are the most common evaluation metrics, as the real-world meaning of mapping error is straightforward in hyperspectral image classification, it can be used as another standard metric for this application.

To conclude, the proposed method MDL4OW achieved state-of-the-art results in terms of all three of the evaluation metrics under the few-shot context and is recommended for hyperspectral image classification. The classwise version should be used only when there are enough training samples.

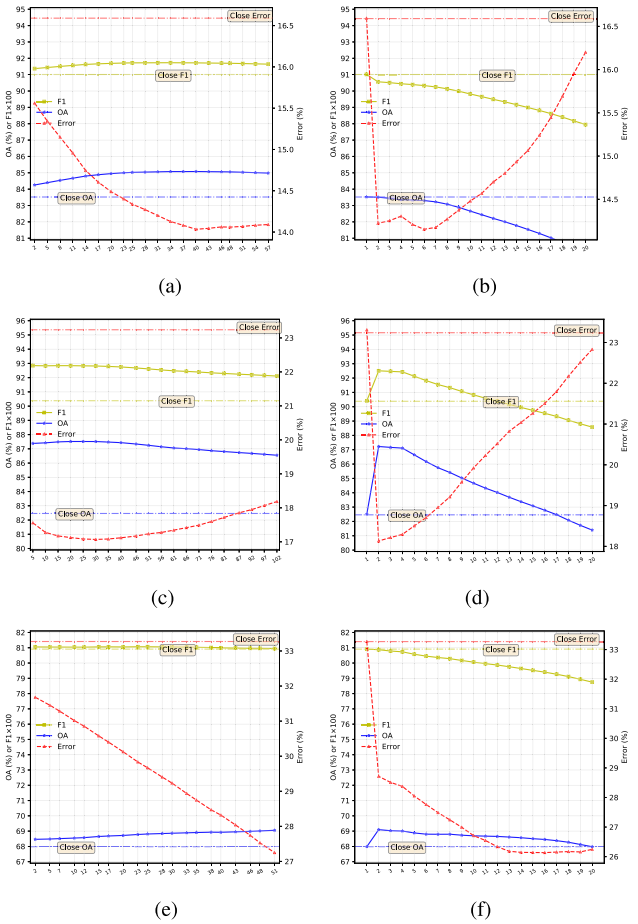


Fig. 10. Sensitivity of the tail number τ. Note that we augmented the training samples in the experiments, so the size of training set should be four times. (a) Pavia, MDL4OW. (b) Pavia, MDL4OW/C. (c) Salinas, MDL4OW. (d) Salinas, MDL4OW/C. (e) Indian, MDL4OW. (f) Indian, MDL4OW/C.

F. Sensitivity of Loss Weights

In this section, we show the sensitivity of loss weights in terms of F1 score and mapping error in Fig. 11 under both few- and many-shot settings. We do not show OA as the OA curve is similar to the F1 score. We set the classification weight from 0.1 to 0.9 and also examine two extreme cases with classification weight as 0.05 and 0.95. The first row to the third row show results on the Pavia, Salinas, and Indian data sets, respectively. In general, both MDL4OW and MDL4OW/C can enhance the classification in terms of F1 score and mapping error, regardless of the loss weights on classification and reconstruction. One exception is Fig. 11(a): we can see that the classwise method MDL4OW/C would suffer from limited training samples. MDL4OW, however, successfully improved the classification in terms of the F1 score.

Although, in general, the methods are insensitive to the losses' weights, when under a few-shot setting, a classification weight smaller than 0.2–0.3 would degrade the performance. The best results are obtained from a classification weight of 0.7–0.8. As for the many-shot setting, the performance is similar for the Pavia and Salinas data. However, for the Indian data, a larger classification weight leads to a decrease in F1 score. Nevertheless, setting the classification

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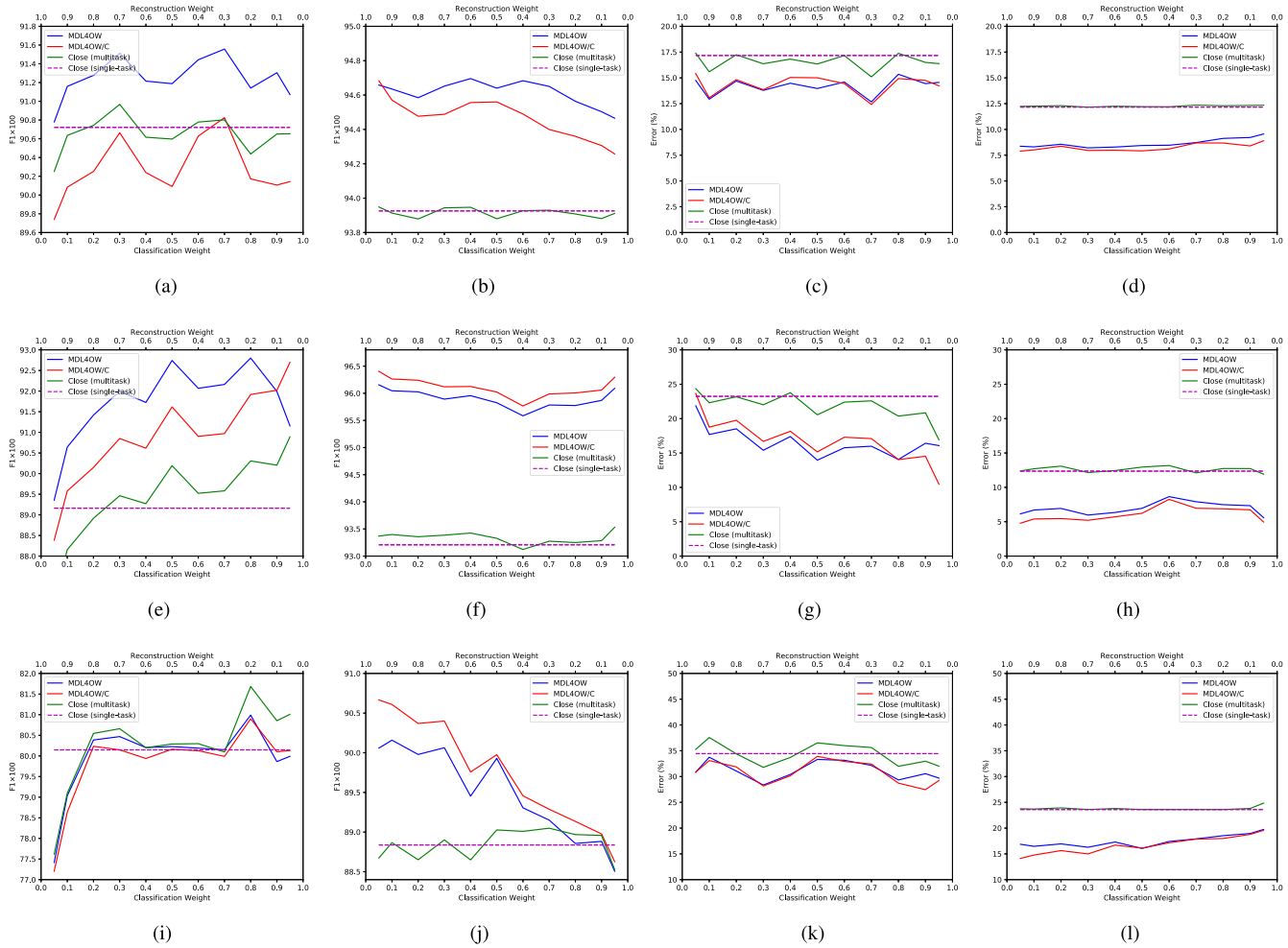


Fig. 11. Sensitivity of loss weights. From the first row to the third row: results on the Pavia, Salinas, and Indian data sets, respectively. From the first column to the fourth column: F1 score (few-shot), F1 score (many-shot), mapping error (few-shot), and mapping error (many-shot). F1 score: the larger the better; mapping error: the smaller the better. (a) Pavia, F1, few-shot. (b) Pavia, F1, many-shot. (c) Pavia, mapping error, few-shot. (d) Pavia, mapping error, many-shot. (e) Salinas, F1, few-shot. (f) Salinas, F1, many-shot. (g) Salinas, mapping error, few-shot. (h) Salinas, mapping error, many-shot. (i) Indian, F1, few-shot. (j) Indian, F1, many-shot. (k) Indian, mapping error, few-shot. (l) Indian, mapping error, many-shot.

TABLE XVIII
COMPUTATION TIME IN SECONDS

	Pavia		Salinas		Indian	
	Train	Predict	Train	Predict	Train	Predict
DCCNN [29]	68	19	116	11	123	2
WCRN [73]	32	13	62	9	65	1
HResNet [43]	36	19	74	16	73	3
CROSR [52]	53	41	90	38	80	12
MDL4OW	61	33	108	30	91	8
MDL4OW/C		34		31		8

and reconstruction weights as 0.3–0.7 can guarantee an acceptable performance of the proposed method.

G. Computation Time

Finally, we show the computation time on the three data sets in Table XVIII under the few-shot context. Note MDL4OW and MDL4OW/C share the same training procedure and have the same training time. The training and predicting times of MDL4OW and MDL4OW/C are longer than its baseline, HResNet. This is because the added reconstruction task requires additional computation. As for CROSR, although the network used is similar to MDL4OW and MDL4OW/C, it does not compute the reconstruction error and, thus, has a lower training time. However, since it calculates the centroid to estimate unknown scores, additional computation is required during the predicting phase.

V. CONCLUSION

In this article, we proposed a novel multitask deep learning method for hyperspectral image classification with unknown classes in the open world, which is a first in the literature to the best of our knowledge. Conventional hyperspectral image classification assumes that the classification system is complete, and there will be no unknown classes in the unseen data. This assumption is risky in the real world since the Earth's surface is complex, and it is difficult to construct a perfect classification system, including all the potential classes, as shown in this study using the popular hyperspectral benchmarks as examples. In this case, the number of known classes will be overestimated, resulting in an overestimation of certain land cover, e.g., crop area for food production.

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To tackle this issue, a multitask deep learning method named MDL4OW is proposed for hyperspectral image classification with unknown classes, where a multitask network is utilized to simultaneously conduct classification and reconstruction. The classification provides the probability of known classes, whereas reconstruction is used to estimate the unknown score. Extensive experiments showed that the proposed method could significantly improve the classification accuracy in terms of OA, F1, and mapping error compared with the state-of-the-art methods currently used, especially when the training samples are scarce.

Further analysis showed that the completeness of a classification system plays an important role in hyperspectral image classification with unknown classes. If a data set is lacking basic land cover elements (impervious, vegetation, and water), the proposed method will be extremely useful to reject the unknown classes and increase the classification accuracy. However, if the classification system includes all basic land cover elements, it will be difficult to identify the unknown classes. Future works should focus on this aspect to provide more accurate hyperspectral image classification. In addition, the use of EVT is empirical. How to properly estimate the tail distribution is essential in classification with unknown classes. Future studies may focus on this aspect and utilize Gaussian distribution, skew-normal distribution, and others to obtain a better classification in the open world. The code and annotations used in this article are available at https://sjliu.me/MDL4OW.

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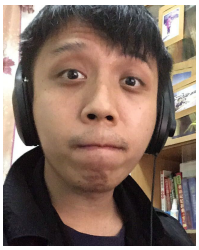
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